Notes: Bernstein, Giroud, and Townsend (JF, 2016)

The Impact of Venture Capital Monitoring

Contents

1 Big picture

- **Question.** Do venture capitalists causally improve the innovation and success of their portfolio companies through postinvestment monitoring, or do they mainly pick firms that would have succeeded anyway?
- **Setting.** The paper studies U.S. venture-backed firms using VentureXpert, patent data, and airline-route data. The core unit is a VC–portfolio-company pair observed over time.
- **Core challenge.** VC-backed firms may outperform because VCs *screen* well rather than because VCs *monitor* well. Comparing VC-backed to non-VC-backed firms does not solve that problem.
- **Main conceptual contribution.** The paper holds selection fixed and studies whether VC involvement changes *after the investment is already made*. The source of variation is the introduction of new airline routes that reduce travel time between a VC and an existing portfolio company.
- **Main empirical design.** Estimate a difference-in-differences regression at the VC–company–year level with pair fixed effects and MSA-by-year fixed effects for both the company location and the VC location.
- **Main baseline result.** When a new airline route lowers travel time, the portfolio company files more patents, receives more citations per patent, and becomes more likely to exit successfully through IPO or acquisition.
- **Magnitudes.** In the baseline specification, the treatment raises patents by about 3.1% and citations per patent by about 5.8%. It raises the IPO probability by about 1.0 percentage point and the probability of IPO or significant acquisition by about 1.4 percentage points.
- **Main mechanism evidence.** The survey evidence says that direct flights lead VCs to visit more often, advise more effectively, and better understand company conditions. In the regression evidence, the effect is concentrated in routes that connect the company to its *lead* VC rather than to nonlead syndicate members.
- **Main credibility result.** The estimates survive controls for local shocks, dynamic pretrend checks, hub-opening and airline-merger treatments, distance-matched control groups, heterogeneous time trends, and several alternative patent measures.
- **Economic takeaway.** VC monitoring matters. Lower monitoring costs improve real outcomes—innovation and exit success—so geography and travel frictions affect how much value VCs create after financing.

2 Data and measurement (key objects)

2.1 Why this setting is useful

- The paper needs a setting where many VC–company relationships are geographically dispersed, so that travel time is economically meaningful.
- This condition is satisfied in the data. The median VC–company distance is about 200 miles, and roughly 40% of portfolio companies are located more than 500 miles from their lead VC.

- VC activity is not confined to Silicon Valley and a few elite clusters. Roughly half of venture-backed companies and VC firms are located outside Northern California, New England, and the New York Tri-State region.
- Because a large number of investments are not local, reductions in travel time can plausibly change how much time VCs spend on-site.

Interpretation. The paper is not exploiting a niche corner of the VC market. It studies a setting where nonlocal monitoring is common enough that airline connections can matter in equilibrium.

2.2 Data sources and sample construction

- **Venture financing data:** Thomson Reuters VentureXpert. This provides funding rounds, investors, company outcomes, and locations.
- **Innovation data:** NBER Patent Data Project, supplemented in some cases with Google Patents citation information.
- **Airline data:** T-100 Domestic Segment Database for 1990–2006 and ER-586 Service Segment Data for 1977–1989, both from the U.S. Department of Transportation.
- **Acquisition values:** SDC Platinum and CapitalIQ, used to separate meaningful acquisitions from small or distressed sales.

Sample construction:

- Keep U.S.-based companies whose first observed financing round is coded as venture stage.
- Baseline sample focuses on **lead VC–portfolio company pairs** because the lead VC is the investor most likely to monitor actively.
- After combining the data sets, the authors observe 22,986 companies funded by 3,158 lead VC firms.

Important data caveats:

- VentureXpert records one main-office location per VC firm, even though some large VC firms have multiple offices.
- Portfolio companies are also observed at one latest-known location.

Interpretation. These data are unusually well suited to the question because they combine *who financed whom*, *where they are*, *how innovation evolves*, and *when travel frictions changed*.

2.3 Treatment: travel-time reductions from new airline routes

The paper defines a treatment when a new airline route reduces the travel time between a VC and its existing portfolio company.

- Travel time is computed from the VC ZIP code to the company ZIP code using the optimal itinerary and mode of transportation (car or airplane).

- The treatment indicator equals one from the year onward in which a new airline route lowers that travel time.
- There are 1,131 treated VC–company pairs in the sample.
- The average travel time reduction is 126 minutes round trip.

Why this is economically meaningful:

- The 126-minute estimate is likely a lower bound because it ignores the compounding probability of delays and missed connections on indirect flights.
- A direct flight may convert an overnight trip into a same-day visit, which is exactly the sort of margin likely to affect monitoring intensity.

Interpretation. The treatment is not "distance" in a loose sense. It is a direct reduction in the time cost of face-to-face monitoring.

2.4 Innovation outcomes

The paper studies both the *quantity* and *quality* of innovation.

- **Patent count:** the number of eventually granted patent applications filed by the company during the year.
- **Citations per patent:** citations received per patent within a three-year window after the grant date.

Measurement details:

- Patent citations proxy for quality and economic importance.
- To deal with truncation at the end of the sample, the baseline uses a three-year citation window after grant.
- The paper also checks robustness using a citation-lag correction and using normalized citation counts by year and technology class.
- Both patent count and citation variables are transformed as $\log(1 + x)$.

Interpretation. The paper is careful not to treat "more patents" as the only outcome. It separately checks whether the patents are also more influential.

2.5 Success outcomes and lead-investor definition

Beyond innovation, the paper asks whether better monitoring leads to *successful exits*.

- **IPO:** indicator equal to one if the company goes public in that year.
- **Success:** indicator equal to one if the company goes public or is acquired for more than $25 million (in 2000 dollars) during that year.

The paper excludes small acquisitions from the success measure because not every acquisition is a good outcome for founders or investors.

Lead-investor definition:

- The lead VC is the investor that has invested in the company the longest.
- This is motivated by prior evidence that the originating VC tends to take the board seat first and is most involved in company decisions.
- Results are robust to alternative lead-VC definitions such as investor with the largest round participation.

Interpretation. The exit outcomes let the paper move beyond innovative activity and ask whether monitoring affects commercially meaningful success.

2.6 Controls and descriptive facts

The baseline control vector includes:

- company age, measured as days since the first financing round;
- a set of indicators for the company's stage of development using VentureXpert's eight-stage classification.

A few descriptive facts matter for interpretation:

- Treated pairs are farther apart than never-treated pairs by construction.
- Treated companies tend to receive less funding and are somewhat less innovative on average.
- Treated VCs are more experienced and more diversified.

Interpretation. These differences matter for comparability, which is why the pair fixed effects and the staggered-treatment design are central: they absorb all time-invariant differences between treated and untreated pairs.

3 Models and empirical specifications (all equations + interpretation)

3.1 Economic mechanism and hypotheses

The paper's logic is very clean:

- If VC-backed firms succeed only because VCs pick good firms at the outset, then changing travel time *after* investment should not matter.
- If VCs add value through monitoring, advice, relationship building, and information acquisition, then lowering travel time should improve company outcomes.

This yields several empirical predictions:

- innovation and exit success should rise after travel time falls;
- the effect should be stronger for the *lead* VC than for passive syndicate members;
- the effect may be stronger for larger travel-time reductions and possibly for earlier-stage firms.

Interpretation. This is a post-selection test of monitoring. The paper is not comparing firms that got VC with firms that did not; it is comparing the same VC–company pair before and after monitoring becomes cheaper.

3.2 Baseline difference-in-differences regression

The main specification is

$$y_{ijt} = \beta \cdot \text{Treatment}_{ijt} + \gamma' X_{ijt} + \alpha_{ij} + \alpha_{\text{MSA}(i)\times t} + \alpha_{\text{MSA}(j)\times t} + \varepsilon_{ijt},$$

where:

- i indexes the portfolio company,
- j indexes the VC firm,
- t indexes the year,
- y_{ijt} is the outcome of interest (patents, citations per patent, IPO, or success),
- X_{ijt} includes company age and financing-stage controls,
- α_{ij} are VC–company pair fixed effects,
- $\alpha_{\text{MSA}(i)\times t}$ are company-location MSA-by-year fixed effects,
- $\alpha_{\text{MSA}(j)\times t}$ are VC-location MSA-by-year fixed effects.

Interpretation of β. The treatment coefficient measures how outcomes change within a given VC–company relationship after a route reduces travel time, relative to the contemporaneous evolution of comparable untreated pairs.

3.3 Panel structure and estimation choice

- The observation is a VC–portfolio company pair by year.
- The treatment is staggered over time, so pairs that are eventually treated serve as controls before the treatment arrives.
- Pair fixed effects absorb permanent pair characteristics such as baseline distance, match quality, and any stable differences between a particular VC and company.

Why this matters:

- Treated and untreated pairs are not identical in levels.
- The empirical design does not rely on that equality. It relies on within-pair changes over time.

3.4 Fixed effects and what each one buys you

The fixed effects are the heart of identification.

- **Pair fixed effects** absorb time-invariant pair differences.
- **Company-MSA $\times$ year fixed effects** absorb local shocks where the portfolio company is located.
- **VC-MSA $\times$ year fixed effects** absorb local shocks where the VC is located.

This is unusually strong for a difference-in-differences design because the treatment is defined at the *pair* level, not just at the region level.

Interpretation. A boom in Salt Lake City that affects all companies there is absorbed. A boom in Cambridge that affects VC firms there is also absorbed. What remains is very specific variation in monitoring costs between a particular VC and a particular firm.

3.5 Standard errors and timing

- Standard errors are clustered at the portfolio-company level to allow serial dependence in outcomes within the same company.
- The paper also checks two-way clustering by company and VC, and the inference barely changes.

Interpretation. The significance is not driven by naive standard-error assumptions.

3.6 Dynamic/event-study specification

To test for pretrends and timing, the paper replaces the treatment dummy with indicators for:

- the year before treatment, Treatment(-1);
- the treatment year, Treatment(0);
- one year after treatment, Treatment(1);
- and two or more years after treatment, Treatment(2+).

Main finding:

- the pre-treatment coefficient is small and insignificant;
- the effect is small in the treatment year;
- it becomes economically large after 12–24 months and remains persistent thereafter.

Interpretation. This timing is exactly what one would expect if lower travel time increases monitoring effort and that better monitoring then translates gradually into patents and exits.

3.7 Mechanism and heterogeneity tests

The paper adds three especially useful mechanism checks.

- **Survey evidence:** direct flights increase travel, flexibility, understanding, and value-add.
- **Lead vs. nonlead VCs:** the effect is concentrated in routes that connect the company to its lead VC.
- **Intensity of treatment:** larger travel-time reductions tend to generate stronger effects.

There is also suggestive appendix evidence that:

- earlier-stage companies respond more strongly;
- the effect is smaller when another VC is already nearby.

Interpretation. The pattern is hard to reconcile with a generic "city got a new flight" story. It looks much more like a monitoring story.

4 Identification and instruments (what makes the estimates credible)

4.1 What identifies the baseline estimate

Identification comes from changes in travel time within an already-existing VC–company relationship.

- The airline route shock is external to the firm's original selection into VC financing.
- The treatment occurs *after* financing is in place.
- The regression asks whether company outcomes improve after monitoring becomes cheaper.

Interpretation. This is the key reason the paper can speak about monitoring rather than screening.

4.2 Why screening is held fixed

The paper's most important conceptual argument is:

- if VC value came only from screening, then changing postinvestment travel costs should have no effect;
- if VC value also comes from monitoring, then lower travel costs should improve outcomes.

Interpretation. This is not a perfect experiment, but it is much closer to a monitoring experiment than the standard VC-backed versus non-VC-backed comparison.

4.3 Local shocks and common unobservables

A major threat is that booming local economies might both attract new airline routes and improve company outcomes.

The paper deals with this directly:

- company-location MSA-by-year fixed effects absorb local shocks where the company operates;
- VC-location MSA-by-year fixed effects absorb local shocks where the VC is based.

The paper also shows the results are similar when local shocks are allowed to be *industry specific*, using MSA-by-industry-by-year fixed effects.

Interpretation. This rules out the simple story that the estimates are just picking up local business cycles.

4.4 Pair-specific shocks and lobbying

A subtler threat is that a shock specific to one VC–company pair could both improve outcomes and trigger lobbying for a direct flight.

The paper argues this is unlikely and provides evidence:

- VC firms and portfolio companies are small relative to airlines and are unlikely to induce route creation on their own;
- there is no evidence of pre-existing treatment effects in the event-study specification;
- the results remain when treatments are restricted to routes created through airline hub openings or mergers, which are harder to attribute to pair-specific lobbying.

Interpretation. Even if lobbying is possible in principle, the data pattern does not look like that story.

4.5 Control-group comparability

Treated pairs differ from never-treated pairs in observable ways, especially distance.

The paper addresses this in several ways:

- pair fixed effects absorb all time-invariant differences;
- the staggered timing means eventually treated pairs are in the control group until treatment;
- the estimates are robust when the control group is restricted to distance-matched pairs;
- the estimates are also robust when using only the sample of eventually treated pairs.

Interpretation. The identification is not relying on long-distance and short-distance pairs being identical in levels.

4.6 Additional robustness checks

The paper runs a long list of further checks:

- interact baseline characteristics with year fixed effects to allow heterogeneous trends;
- use alternative citation measures and truncation corrections;
- cluster standard errors in two dimensions;

- test whether treatment merely improves access to *non-VC* resources in the VC's MSA; it does not.

Interpretation. The results are not fragile to reasonable alternative specifications.

4.7 What the paper can and cannot distinguish

The paper is honest about an important limitation:

- it does *not* directly observe VC monitoring time or a clean intensity measure of on-site involvement;
- therefore it cannot perfectly distinguish among monitoring, advising, relationship building, better information acquisition, and other proximity-based channels.

What it *can* show is:

- lower travel time changes outcomes after financing is already fixed;
- the effect is concentrated where monitoring should matter most;
- the survey evidence is consistent with more in-person involvement.

Interpretation. The paper identifies the causal effect of lower monitoring *costs*, not a structurally separated elasticity of each individual VC activity.

4.8 Relation to the literature

The paper contributes by improving on two common approaches:

- **VC-backed vs. non-VC-backed comparisons:** those mix screening and monitoring.
- **Structural decompositions:** those require stronger modeling assumptions.

Its main contribution is reduced-form causal evidence that postinvestment VC involvement matters.

5 Tables and figures

5.1 Table I and Figures 1–2: sample geography and distance

Read.

- Venture-backed companies and VC firms are geographically dispersed.
- Many states have low shares of portfolio companies funded by a lead VC from the same state.
- The distance distribution has a large local mass, but it also has a long right tail: the median is about 200 miles and around 40% of companies are more than 500 miles from their lead VC.

Economic message. The paper is not identified off a handful of exotic remote matches. Many VC relationships are far enough apart that air travel genuinely matters.

Table I

Sample Composition

This table shows the composition of portfolio companies and VC firms in the sample. Portfolio companies are categorized as "Never Treated" if they never experienced a reduction in travel time to their lead VC investor, and "Ever Treated" otherwise. Similarly, VC firms are categorized as "Never Treated" if they never experienced a reduction in travel time to any of the companies in their portfolio (for which they were a lead investor), and "Ever Treated" otherwise. Panel A shows the company region distribution. Panel B shows the company industry distribution. Panel C shows the VC region distribution.

	Never Treated		Ever Treated		All	
	Freq	Percent	Freq	Percent	Freq	Percent
Panel A: Company Region						
Alaska/Hawaii	22	0.10	1	0.09	23	0.10
Great Lakes	1,054	4.81	51	4.66	1,105	4.81
Great Plains	738	3.37	44	4.02	782	3.40
Mid-Atlantic	1,178	5.38	59	5.39	1,237	5.38
N. California	5,464	24.96	146	13.35	5,610	24.41
New England	2,529	11.55	115	10.51	2,644	11.50
New York Tri-State	2,355	10.76	90	8.23	2,445	10.64
Northwest	854	3.90	48	4.39	902	3.92
Ohio Valley	1,169	5.34	59	5.39	1,228	5.34
Rocky Mountains	875	4.00	44	4.02	919	4.00
S. California	1,980	9.04	120	10.97	2,100	9.14
South	432	1.97	67	6.12	499	2.17
Southeast	1,475	6.74	121	11.06	1,596	6.94
Southwest	1,740	7.95	129	11.79	1,869	8.13
U.S. Territories	27	0.12	0	0	27	0.12
Total	21,892	100.00	1,094	100.00	22,986	100.00
Panel B: Company Industry						
Biotechnology	1,221	5.58	70	6.40	1,291	5.62
Communications and Media	2,243	10.25	109	9.96	2,352	10.23
Computer Hardware	1,307	5.97	75	6.86	1,382	6.01
Computer Software and Services	4,526	20.67	192	17.55	4,718	20.53
Consumer Related	1,428	6.52	91	8.32	1,519	6.61
Industrial/Energy	1,222	5.58	77	7.04	1,299	5.65
Internet Specific	4,137	18.90	135	12.34	4,272	18.59
Medical/Health	2,329	10.64	144	13.16	2,473	10.76
Other Products	1,955	8.93	124	11.33	2,079	9.04
Semiconductors/Other Elect.	1,524	6.96	77	7.04	1,601	6.97
Total	21,892	100.00	1,094	100.00	22,986	100.00
Panel C: VC Region						
Alaska/Hawaii	4	0.15	0	0	4	0.13
Great Lakes	174	6.65	38	7.04	212	6.71
Great Plains	90	3.44	29	5.37	119	3.77
Mid-Atlantic	126	4.81	34	6.30	160	5.07
N. California	502	19.17	60	11.11	562	17.80

(Continued)

Table I—*Continued*

Panel C: VC Region	Never Treated		Ever Treated		All	
	Freq	Percent	Freq	Percent	Freq	Percent
New England	210	8.02	84	15.56	294	9.31
New York Tri-State	615	23.49	129	23.89	744	23.56
Northwest	67	2.56	9	1.67	76	2.41
Ohio Valley	143	5.46	34	6.30	177	5.60
Rocky Mountains	82	3.13	13	2.41	95	3.01
S. California	204	7.79	27	5.00	231	7.31
South	58	2.22	20	3.70	78	2.47
Southeast	145	5.54	26	4.81	171	5.41
Southwest	196	7.49	37	6.85	233	7.38
U.S. Territories	2	0.08	0	0	2	0.06
Total	2,618	100.00	540	100.00	3,158	100.00

5.2 Table III and Figure 3: survey evidence

Read.

- The survey includes 306 respondents from the VC industry.
- About 77% are partners.
- Respondents report an average portfolio size of 6.58 companies, 3.79 of which are local, and an average of 9.93 visits per company per year.
- They report spending 48% of their time monitoring and assisting portfolio companies.
- 86% agree that direct flights increase the time VCs spend in person at portfolio companies.
- 83% agree that direct flights let VCs advise companies more effectively.
- Around 80% agree that direct flights help VCs add value and better understand company challenges.
- In the specific scenario questions, 83% agree that a direct flight increases travel frequency and 89% agree that it increases flexibility to travel when useful.

Economic message. The survey does not identify the causal effect, but it strongly supports the paper's core mechanism assumption: lower travel time increases on-site VC involvement.

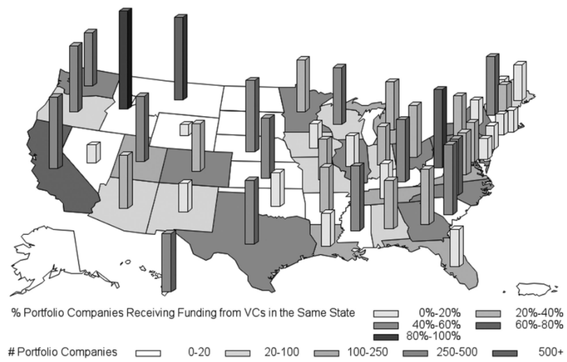

Figure 1. VC-portfolio company pairs. This figure shows the distribution of portfolio companies across states. Darker states are those with more portfolio companies. The height of the bars indicates the percentage of companies funded by a lead VC in the same state.

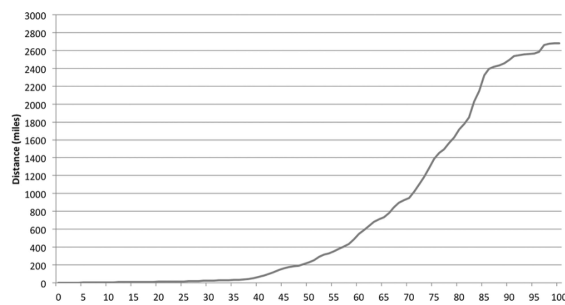

Figure 2. CDF of distance distribution. This figure plots the cumulative density function (CDF) of the VC-portfolio company distance distribution.

Table III

Survey Evidence

This table shows the results of a survey of VC investors. Panel A shows the distribution of respondents across countries and U.S. states. Panel B summarizes the responses to the preliminary questions. Panels C and D summarize the responses to the key questions shown in Section VI of the Internet Appendix. The questions in Panel C pertain to general VC behavior, whereas the questions in Panel D correspond to the behavior of the respondents. On all questions, a standard six-point Likert scale is used, where potential responses range from Strongly Disagree (= 1) to Strongly Agree (= 6). The % Agree column represents the percent of respondents that somewhat agreed, agreed, or strongly agreed with the statement. A t-test is done to determine if the mean response is statistically different from the neutral midpoint of 3.5. * , **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

Panel A: Geographical Distribution

Nation	Percent	State	Percent
Brazil	2.26	California	69.52
Canada	0.75	Connecticut	0.95
China	2.75	Illinois	0.95
Germany	3.76	Louisiana	0.95
Hong Kong	1.5	Maryland	0.95
Israel	2.26	Massachusetts	13.33
Japan	0.75	New Hampshire	0.95
Poland	0.75	New Mexico	0.95
Portugal	0.75	New York	4.76
Russia	0.75	Pennsylvania	0.95
Singapore	0.75	Texas	2.86
South Africa	1.5	Utah	0.95
South Korea	1.5	Virginia	0.95
Sweden	0.75	Washington	0.95
Switzerland	0.75		
U.S.	78.95		

Panel B: Preliminary Questions

	N	Mean	Median	St. Dev.
Partner (0/1)	306	0.77	1	0.42
Currently a VC (0/1)	306	0.61	1	0.49
Years since last worked in VC	113	8.65	6	7.82
Number of companies	306	6.58	5	3.98
Number of local companies	304	3.79	3	3.52
Number of visits per year	303	9.93	6	12.6
Visit local companies more than nonlocal (0/1)	303	0.71	1	0.46
Percent of time spent monitoring	306	0.48	0.50	0.18

(Continued)

Table III—*Continued*

Panel C: General Questions

Direct flights allow VCs to . . .	*N*	% Agree	Mean	St. Dev.
Spend more time assisting/monitoring in person	306	0.86	4.43***	1.18
More effectively advise companies	306	0.83	4.31***	1.20
Add more value to companies	306	0.80	4.20***	1.27
Better understand key challenges/issues facing companies	306	0.80	4.32***	1.31

Panel D: Specific Questions

The introduction of a direct flight will . . .	*N*	% Agree	Mean	St. Dev.
Increase frequency of travel	306	0.83	4.28***	1.14
Increase flexibility to travel when useful	306	0.89	4.74***	1.13
Help communicate more effectively	306	0.72	3.90***	1.22
Help establish better relationships	306	0.81	4.31***	1.20
Help add more value	306	0.75	3.93***	1.20
Help understand state of company	306	0.76	4.16***	1.22

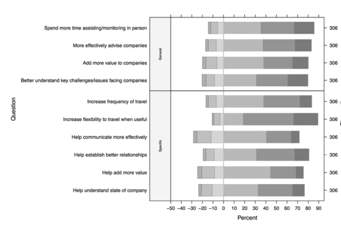

Figure 3. Survey responses. This figure shows the distributions of responses to the survey questions provided in Section VI of the Internet Appendix. The first four rows show the responses to the general questions, while the last six rows show the responses to the specific questions. On the horizontal axis, positive (negative) percentages refer to "agree" ("disagree") responses.

5.3 Table IV: main regressions

Read.

- In the most demanding specification, treatment raises patents by 3.1%.
- It raises citations per patent by 5.75%.
- It raises the IPO probability by roughly 1.04 percentage points.
- It raises the probability of IPO or significant acquisition by roughly 1.35 percentage points.
- These estimates survive the addition of controls and the two sets of MSA-by-year fixed effects.

Economic message. Lower monitoring costs affect both innovative output and commercially meaningful success. This is the main causal evidence of the paper.

Table IV

Main Regressions

This table shows the main results. Observations are at the company-VC pair by year level. Only pairs involving a lead investor are included in the sample. *Treatment* is an indicator variable equal to one if a new airline route that reduces the travel time between the VC and the portfolio company has been introduced. *Patents* is equal to the log of (one plus) the number of patents the portfolio company applied for during the year. *Citations/Patent* is equal to the log of (one plus) the number of citations those patents received (in the three years following their grant date) divided by the number of patents. *IPO* is an indicator variable equal to one if the company went public that year. *Success* is an indicator variable equal to one if the company went public or was acquired (for over $25 million in 2000 dollars) that year. Standard errors, clustered by portfolio company, are shown in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

Panel A: Innovation

	Patents			Citations/Patent		
	(1)	(2)	(3)	(4)	(5)	(6)
Treatment	0.0371***	0.0352***	0.0310***	0.0744***	0.0698***	0.0575***
	(0.00975)	(0.00971)	(0.0113)	(0.0178)	(0.0178)	(0.0203)
Controls	No	Yes	Yes	No	Yes	Yes
Pair FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	No	Yes	Yes	No
MSA (VC) × Year FE	No	No	Yes	No	No	Yes
MSA (Company) × Year FE	No	No	Yes	No	No	Yes
R^2	0.638	0.640	0.668	0.546	0.547	0.576
Observations	130,169	130,169	130,169	130,169	130,169	130,169

Panel B: Exits

	IPO			Success		
	(1)	(2)	(3)	(4)	(5)	(6)
Treatment	0.0103***	0.00994***	0.0104**	0.0113**	0.0112**	0.0135**
	(0.00378)	(0.00373)	(0.00429)	(0.00507)	(0.00493)	(0.00577)
Controls	No	Yes	Yes	No	Yes	Yes
Pair FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	No	Yes	Yes	No
MSA (VC) × Year FE	No	No	Yes	No	No	Yes
MSA (Company) × Year FE	No	No	Yes	No	No	Yes
R^2	0.435	0.440	0.494	0.399	0.405	0.453
Observations	130,169	130,169	130,169	130,169	130,169	130,169

Table V

Dynamics

This table shows the dynamics of the treatment effects. All variables are defined as in Table IV. The variable *Treatment(-1)* is an indicator variable equal to one if the observation is recorded in the year preceding the treatment. *Treatment(0)*, *Treatment(1)*, and *Treatment(2+)* are defined analogously with respect to the year of the treatment, the first year after the treatment, and two or more years after the treatment, respectively. Standard errors, clustered by portfolio company, are shown in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

	(1) Patents	(2) Citations/Patent	(3) IPO	(4) Success
Treatment(−1)	0.00639	0.0170		
	(0.0147)	(0.0285)		
Treatment(0)	0.0165	0.0244	0.00682	0.0114
	(0.0155)	(0.0283)	(0.00502)	(0.00710)
Treatment(1)	0.0391**	0.0690**	0.00805	0.0110
	(0.0182)	(0.0333)	(0.00644)	(0.00842)
Treatment(2+)	0.0494***	0.106***	0.0158**	0.0172**
	(0.0182)	(0.0326)	(0.00655)	(0.00831)
Controls	Yes	Yes	Yes	Yes
Pair FE	Yes	Yes	Yes	Yes
MSA (VC) × Year FE	Yes	Yes	Yes	Yes
MSA (Company) × Year FE	Yes	Yes	Yes	Yes
R^2	0.668	0.576	0.494	0.453
Observations	130,169	130,169	130,169	130,169

5.4 Table V: dynamics of the treatment

Read.

- The coefficient on Treatment(−1) is small and insignificant.
- The treatment year itself shows only a small effect.
- The economically important effects appear one year later and persist into year 2 and beyond.

Economic message. The absence of pretrends strengthens identification, while the delayed response is exactly what one would expect if lower travel costs change real company performance through monitoring.

5.5 Table VI: lead versus nonlead VCs

Read.

- When the treatment is defined using a *nonlead* investor in a different MSA from the lead VC, the coefficients are close to zero and insignificant for patents, citations, IPO, and success.
- The cities connected by lead and nonlead treatments look broadly similar in terms of population, income, and geography.

Economic message. The effect is not "any new route to any investor." It is specifically about the investor who is expected to do the monitoring.

Table VI
Nonlead VCs

Panel A repeats the analysis of Table IV, but restricts the sample to company-VC pairs that do not involve a lead investor. Panel B compares mean VC MSA characteristics (in the treatment year) for treatments involving lead and nonlead investors. Nonlead VCs located in the same MSA as the lead VC are excluded from the sample in both panels. Standard errors, clustered by portfolio company, are shown in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

Panel A: Effect of Nonlead Treatment

	(1) Patents	(2) Citations/Patent	(3) IPO	(4) Success
Treatment	−0.0128	−0.0205	0.00761	0.0139
	(0.0203)	(0.0368)	(0.00691)	(0.00972)
Controls	Yes	Yes	Yes	Yes
Pair FE	Yes	Yes	Yes	Yes
MSA (VC) × Year FE	Yes	Yes	Yes	Yes
MSA (Company) × Year FE	Yes	Yes	Yes	Yes
R^2	0.758	0.688	0.673	0.627
Observations	90,609	90,609	90,609	90,609

Panel B: Lead vs. Nonlead Treatment Characteristics

	Lead Treat		Nonlead Treat		Difference	
	Mean	Std. Dev.	Mean	Std. Dev.	Mean	Std. Err.
VC MSA income (Billions)	161.1	201.4	150.6	190.1	10.5	8.53
VC MSA population (Millions)	4.85	5.56	4.49	5.18	0.36	0.23
VC MSA income per capita (Thousands)	33.1	12.1	33.5	13.1	−0.41	0.55
VC in Northern California	0.087	0.28	0.10	0.30	−0.014	0.012
VC in New York Tri-State	0.22	0.42	0.20	0.40	0.018	0.017
VC in New England	0.19	0.39	0.19	0.40	−0.0037	0.017
Observations	1,131		1,068		2,199	

Table VII
Intensity of the Treatment

This table repeats the analysis of Table IV, but separates the treatment indicator into two variables. *Treatment × Large* is an indicator variable equal to one if the treatment is associated with a travel time reduction of at least 60 minutes, while *Treatment × Small* is an indicator variable equal to one if the treatment is associated with a travel time reduction of less than 60 minutes. Standard errors, clustered by portfolio company, are shown in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

	(1) Patents	(2) Citations/Patent	(3) IPO	(4) Success
Treatment × Large	0.0336**	0.0684***	0.0115**	0.0138**
	(0.0143)	(0.0248)	(0.00524)	(0.00701)
Treatment × Small	0.0259	0.0359	0.00822	0.0129
	(0.0173)	(0.0333)	(0.00683)	(0.00948)
Controls	Yes	Yes	Yes	Yes
Pair FE	Yes	Yes	Yes	Yes
MSA (VC) × Year FE	Yes	Yes	Yes	Yes
MSA (Company) × Year FE	Yes	Yes	Yes	Yes
R^2	0.668	0.576	0.494	0.453
Observations	130,169	130,169	130,169	130,169

Table VIII
Robustness

All regressions presented in this table are variants of the baseline specification in Table IV. Panel A separates the treatment indicator into two variables. *Treatment (Hub or Merger)* is an indicator variable equal to one if the treatment is due to the opening of a new airline hub, or the merger of two airlines. *Treatment (Other)* is an indicator variable equal to one if the treatment is not due to a hub opening or merger. Panel B restricts the sample to the eventually treated pairs. Standard errors, clustered by portfolio company, are shown in parentheses. * , **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

Panel A: Hub Openings and Airline Mergers

	(1) Patents	(2) Citations/Patent	(3) IPO	(4) Success
Treatment (Hub or Merger)	0.0540**	0.116**	0.0237*	0.0325*
	(0.0255)	(0.0508)	(0.0142)	(0.0176)
Treatment (Other)	0.0273**	0.0475**	0.00842*	0.0105*
	(0.0126)	(0.0219)	(0.00433)	(0.00593)
Controls	Yes	Yes	Yes	Yes
Pair FE	Yes	Yes	Yes	Yes
MSA (VC) × Year FE	Yes	Yes	Yes	Yes
MSA (Company) × Year FE	Yes	Yes	Yes	Yes
R^2	0.668	0.576	0.494	0.453
Observations	130,169	130,169	130,169	130,169

Panel B: Eventually Treated Pairs

	(1) Patents	(2) Citations/Patent	(3) IPO	(4) Success
Treatment	0.0314***	0.0354*	0.0250***	0.0376***
	(0.0107)	(0.0207)	(0.00414)	(0.00517)
Controls	Yes	Yes	Yes	Yes
Pair FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
R^2	0.582	0.440	0.218	0.211
Observations	7,978	7,978	7,978	7,978

5.6 Tables VII–VIII: treatment intensity and robustness

Read.

- Larger travel-time reductions (at least 60 minutes) produce stronger point estimates than smaller reductions.
- Treatments due to hub openings or airline mergers also show positive and significant effects.
- Using only eventually treated pairs gives similar estimates.

Economic message. The results are not driven by weak travel changes, odd control-group composition, or endogenous route creation specific to a particular pair.

6 Short summary

- The paper asks whether VCs add value through monitoring rather than only through screening.
- It uses new airline routes that reduce travel time between VCs and existing portfolio companies as shocks to monitoring costs.
- In a VC–company pair difference-in-differences design with very demanding fixed effects, lower travel time increases patents, citations per patent, IPOs, and successful exits.

- The effects arrive with a lag, which supports a real mechanism rather than a spurious contemporaneous shock.
- Survey evidence and the lead-versus-nonlead comparison both support the interpretation that lower travel time increases actual VC involvement.
- The core implication is that on-site monitoring by VCs is an important input into start-up innovation and success.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that VC monitoring matters *causally*. By using postinvestment reductions in travel time, it shows that when it becomes easier for VCs to visit companies, innovation and successful exits increase.

7.2 Economic intuition

The basic intuition is simple:

- face-to-face interaction is costly when travel is slow and inconvenient;
- cheaper travel makes active monitoring, advising, and information acquisition more feasible;
- those activities improve start-up decisions and performance.

The evidence that the effect is concentrated in lead VCs and emerges only with a lag fits this interpretation very well.

7.3 Takeaway

For this paper, geography matters not because it changes *which* firms get financed, but because it changes how intensively investors can work with firms *after* financing. That is why the paper is so important: it gives reduced-form evidence that VC value creation is not only about picking winners, but also about helping them become winners.